

Critical Design Review: Vehicle Control Unit Architecture for Commercial Three-Wheeled Electric Vehicles

1. Executive Summary

The transition of the commercial three-wheeled vehicle—often colloquially referred to as the auto-rickshaw—from a conventional internal combustion engine to a fully electric powertrain featuring independent rear hub motors represents a profound paradigm shift in vehicle dynamics. In traditional architectures, a mechanical rear axle and differential inherently govern the equalization of torque, providing mechanical stability and predictable understeer characteristics. The removal of this mechanical linkage in favor of an active, software-defined Vehicle Control Unit (VCU) transfers the entire burden of lateral stability, traction control, and yaw manipulation to the embedded control algorithms. The proposed VCU architecture for this delta-footprint (one front wheel, two independent rear wheels) commercial electric vehicle presents an ambitious integration of Extended Kalman Filtering (EKF), Sliding Mode Control (SMC), and a Predictive Lateral Load Transfer Ratio (PLTR) equation for rollover mitigation. However, a rigorous and exhaustive critical design review of this proposed architecture reveals severe mathematical vulnerabilities, control theory impracticalities, and critical functional safety (ISO 26262) violations. The operational environment of a commercial auto-rickshaw is characterized by extreme constraints. The gross vehicle weight (GVW) fluctuates radically, often more than doubling from an unloaded baseline of 400 kilograms to a fully loaded state exceeding 900 kilograms. This massive variance dynamically shifts the vehicle's center of gravity (CG) both longitudinally and vertically. Furthermore, the vehicle must operate in harsh commercial environments featuring severely degraded road surfaces, asymmetric friction coefficients (μ -split surfaces), and steep gradients, all while relying on standard, cost-constrained commercial-grade sensors rather than high-fidelity aerospace instrumentation.

The core deficiencies within the proposed architecture stem from the direct application of four-wheeled mathematical models to a three-wheeled delta geometry, the deployment of continuous-time high-frequency switching controllers onto hardware possessing massive unsprung inertia, and the creation of catastrophic single points of failure within the state estimation logic. Most critically, the reliance on a single, un-driven front wheel for longitudinal velocity estimation virtually guarantees a total systemic collapse during emergency braking maneuvers. Concurrently, the proposed degradation state machines and hardware failsafe mechanisms rely on legacy industrial concepts that actively violate the stringent requirements of the ISO 26262 automotive safety standard.

This comprehensive report dissects the proposed VCU architecture across five primary

domains: the mathematical and physics boundaries of the kinematic models, the practical execution of the control theory on embedded hardware, the systemic collapse during kinematic edge cases, the legal and functional safety compliance of the fallback states, and the necessary architectural remediation required to achieve commercial viability and regulatory approval.

2. Contextualizing the Delta-Footprint Dynamics and Commercial Constraints

Before evaluating the specific control modules, it is imperative to establish the physical realities and constraints of the target vehicle. The delta-footprint three-wheeler is inherently compromised regarding lateral stability. Unlike a four-wheeled vehicle, which benefits from a rectangular stability footprint, or a tadpole-footprint three-wheeler (two front wheels, one rear wheel), which places the widest track at the primary axis of steering and weight transfer, the delta configuration is uniquely prone to combined directional instability and lateral rollover. When a delta-footprint vehicle initiates a turn, the lateral acceleration generates a centrifugal force acting through the vehicle's center of gravity. Because the front axle consists of a single point of contact, the vehicle does not roll over a longitudinal axis parallel to the chassis centerline. Instead, it rolls over a diagonal tipping axis connecting the front tire contact patch to the outer rear tire contact patch. This fundamental geometric reality dictates that any control logic attempting to govern yaw or predict rollover must account for this diagonal projection of forces.

Compounding this geometric vulnerability is the massive payload variance inherent to commercial auto-rickshaws. In an unloaded state, with only the driver present in the forward cabin, the center of gravity is positioned relatively low and forward. In this state, the vehicle might exhibit mild understeer. However, as the rear passenger cabin is loaded to maximum capacity—potentially with passengers and cargo exceeding 500 kilograms—the center of gravity migrates aggressively rearward and upward. This migration dramatically alters the vehicle's yaw moment of inertia (I_z), decreases the distance from the CG to the rear axle, and elevates the CG height (h), making the vehicle highly susceptible to oversteer and catastrophic un-tripped rollovers.

The VCU must manage these dynamic shifts using an inherently limited sensor suite. The architecture specifies a standard Steering Angle Sensor (SAS), standard wheel speed encoders, and a commercial-grade 6-axis Inertial Measurement Unit (IMU). Commercial MEMS (Micro-Electro-Mechanical Systems) IMUs suffer from significant bias instability, random walk noise, and thermal drift over time. Wheel speed encoders operating on rough, pothole-riddled roads produce high-frequency quantization noise and are subject to localized tire slip-stick phenomena. The VCU algorithms cannot assume access to pristine, idealized signals; they must be aggressively robust against corrupt, delayed, and noisy data streams.

3. Mathematical and Physics Loopholes in the

Proposed Kinematics

The foundational mathematics of the proposed VCU architecture rely heavily on idealized kinematic approximations that fail to capture the asymmetric, dynamic nature of the delta-footprint three-wheeler. By porting standard passenger car equations directly into the control modules, the architecture introduces systemic calculation errors that will inevitably compromise the stability controller in real-world environments.

3.1 The Rollover Dynamics Fallacy: Four-Wheeled PLTR on a Delta Footprint

The proposed architecture utilizes a Predictive Lateral Load Transfer Ratio (PLTR) equation defined as $PLTR \approx \frac{2 \cdot h \cdot a_y}{g \cdot B} + C_p(\dot{\delta}_f \cdot v_x)$ to predict and mitigate rollover events. If the PLTR value exceeds 0.8, the system triggers a danger mode intervention. This equation is fundamentally flawed for the specified vehicle geometry and will result in highly dangerous dynamic behavior.

The expression $\frac{2 \cdot h \cdot a_y}{g \cdot B}$ is the universally recognized static stability factor adjustment for a conventional four-wheeled vehicle. In this equation, h represents the height of the center of gravity, a_y is the lateral acceleration, g is the acceleration due to gravity, and B represents the track width of the vehicle. This formulation assumes that the rollover axis is perfectly parallel to the longitudinal centerline of the chassis, meaning the lateral distance from the center of gravity to the tipping point is always exactly half of the track width ($B/2$). In a delta-footprint three-wheeler, the vehicle does not roll over a longitudinal axis. The tipping axis is a diagonal line drawn from the front tire contact patch to the outer rear tire contact patch, forming a triangular stability footprint.¹ Because the tipping axis is diagonal, the effective track width at the location of the center of gravity is not a constant dimension B . It is mathematically dependent on the longitudinal distance from the front axle to the center of gravity (l_f). Using the principles of geometric similar triangles, the effective half-track width (d_{tip}) that resists the overturning moment is defined as $d_{tip} = \frac{B}{2} \cdot \frac{l_f}{L}$, where L is the total wheelbase of the vehicle.⁴

Because l_f is strictly less than L , the effective lever arm resisting rollover (d_{tip}) in a delta trike is always substantially smaller than $B/2$. By utilizing the unadjusted track width B in the denominator, the proposed VCU equation vastly overestimates the mechanical stability of the vehicle. The controller will calculate a safe PLTR value while the physical vehicle is already in the process of tipping over.

Furthermore, the architecture fails to account for the massive payload variance of the

commercial auto-rickshaw. As the vehicle is loaded from 400 kilograms to 900 kilograms, the center of gravity migrates rearward, increasing l_f , and migrates upward, increasing h . If the VCU utilizes hardcoded, static parameters based on the unloaded vehicle state, it will completely fail to predict an imminent rollover when the vehicle is fully loaded with passengers. Conversely, if it uses the loaded values, it will trigger severe, false-positive rollover interventions—such as commanding massive negative braking torque—when the driver is operating the vehicle alone. The overturning moment is also generated not just by lateral acceleration (a_y), but by the vectorial projection of a_y perpendicular to the diagonal tipping axis, which is heavily compounded by longitudinal acceleration (a_x) or deceleration during trail braking into a corner.⁶ The failure to vectorially project the inertial forces across the true delta stability triangle renders the $PLTR > 0.8$ threshold arbitrary, mathematically unsound, and physically dangerous.

Parameter	Proposed VCU Assumption (Four-Wheeled)	Physical Reality (Delta Three-Wheeler)	Consequence of Error
Tipping Axis	Longitudinal, parallel to centerline.	Diagonal, connecting front to outer rear tire.	Overestimates lateral stability threshold.
Lever Arm	Constant half-track width ($B/2$).	Variable geometry ($d_{tip} = \frac{B}{2} \cdot$).	Controller intervenes too late to prevent un-tripped roll.
CG Position (l_f, h)	Treated as static constants in the equation.	Highly variable (400 kg to 900 kg payload shifts).	False positives when empty; missed predictions when fully loaded.
Overturning Force	Purely lateral acceleration (a_y).	Vector sum of lateral (a_y) and longitudinal (a_x) forces.	Trail braking into corners induces undetected rollover vectors.

3.2 Kinematic Estimation Flaws: Accelerometer Coupling and Gradient Vulnerabilities

The Demand Generation module proposes real-time gradient estimation utilizing the equation $\theta = \arcsin((a_{x,IMU} - \dot{v}_x)/g)$ to enable Uphill Anti-Rollback and Downhill 0% Regen maps. While this equation represents a theoretically sound kinematic relationship in a sterile, academic simulation, it is profoundly fragile when applied to a commercial-grade embedded system operating in a high-vibration environment.

The fundamental mathematical loophole resides in the reliance on the derivative term $\dot{v}_x$. The

architecture calculates $\dot{v}_x$ by taking the numerical derivative of the longitudinal vehicle speed, which is derived from the front wheel speed encoder data. Wheel speed encoders operating over rough, unpaved, or pothole-riddled commercial roads experience continuous micro-stutters, localized tire slip, and suspension hop, resulting in a signal plagued by high-frequency quantization noise.⁸ Taking the direct numerical derivative of a noisy, discrete-time signal amplifies the noise exponentially. Even if the proposed Low-Pass Filter (LPF) is applied, the filtering process introduces severe phase lag. This phase lag causes a temporal mismatch between the instantaneous acceleration measured by the IMU ($a_{x,IMU}$) and the delayed, smoothed derivative of the wheel speed ($\dot{v}_x$). Subtracting two signals that are out of phase results in massive transient spikes in the calculated angle θ , entirely corrupting the gradient estimation.

Moreover, the equation implicitly assumes that the vehicle chassis remains perfectly parallel to the road surface at all times. In reality, a commercial auto-rickshaw utilizes soft suspension tuning to accommodate heavy passenger payloads. During aggressive acceleration, the vehicle experiences longitudinal load transfer, resulting in suspension "squat" (rearward pitch of the chassis). During heavy braking, the vehicle experiences suspension "dive" (forward pitch). Because the IMU is hard-mounted to the chassis structure, the pitch angle of the chassis (θ_{pitch}) relative to the road plane introduces a gravitational component into the a_x reading.

The true measured acceleration of the IMU is mathematically defined as

$a_{x,IMU} = \dot{v}_x + g \cdot \sin(\theta_{road} + \theta_{pitch})$. The proposed VCU logic completely conflates suspension pitch with physical road gradient. Consequently, heavy acceleration on a perfectly flat road will induce chassis squat, which the IMU will register as an incline. The VCU will falsely calculate an uphill gradient and potentially trigger unwarranted Anti-Rollback torque stacking, creating unpredictable and surging longitudinal acceleration that undermines driver intent.¹⁰

3.3 Asymmetric Yaw Generation in Unconstrained Traction Control

The proposed Torque Allocation module incorporates an active traction control logic tailored for μ -split surfaces (e.g., scenarios where the left wheel is on low-friction dirt while the right wheel is on high-friction tarmac). The logic dictates that the VCU calculates the localized tire

slip ratio: $\lambda = \frac{|\omega_{wheel} \cdot r_{tire} - v_x|}{\max(|\omega_{wheel} \cdot r_{tire}|, |v_x|)}$. If λ exceeds a 15% threshold, the VCU throttles the slipping tire while maintaining full torque on the gripping side.

This logic presents a critical physics loophole in the context of independent rear hub motors.

When a conventional vehicle with an open differential accelerates on a μ -split surface, the differential physically limits the maximum torque applied to the gripping wheel to match the torque capacity of the slipping wheel, inherently maintaining symmetric, straight-line acceleration (albeit at a much lower overall magnitude). However, independent rear hub motors are mechanically decoupled.

If the proposed VCU aggressively reduces torque to the left wheel to arrest longitudinal slip, but permits the right hub motor to execute the maximum driver-demanded torque, it generates a massive, instantaneous torque differential ($\Delta T_{asym} = T_{Right} - T_{Left}$). This torque differential acts as a direct yaw moment vector ($M_z = \Delta T_{asym} \cdot \frac{B_{eff}}{2 \cdot r_{tire}}$) around the vehicle's vertical axis.¹² The vehicle will violently yaw and pull toward the high-grip side, an effect known as severe torque steer. In a three-wheeler with a single front wheel, the front tire has an inherently limited lateral force capacity (cornering stiffness) to counteract this sudden, massive rear-induced yaw moment. The proposed traction control logic focuses entirely on longitudinal slip optimization without establishing an overarching yaw-moment boundary constraint, virtually guaranteeing catastrophic lateral instability and spin-outs during aggressive acceleration on mixed-friction surfaces.

4. Control Theory Practicality and Embedded Hardware Feasibility

The translation of theoretical control laws into physical actuation introduces a myriad of real-world complexities, including system latencies, quantization errors, and electromechanical dynamic coupling. The proposed architecture relies heavily on Sliding Mode Control (SMC) for yaw stability and an Extended Kalman Filter (EKF) for state estimation. Both of these control strategies present severe, prohibitive practical challenges when mapped to commercial-grade microcontroller hardware and direct-drive in-wheel motor topologies.

4.1 Sliding Mode Control Chattering and Hub Motor Unsprung Mass Resonance

The Yaw Control module utilizes a Sliding Mode Controller to calculate the corrective yaw moment (ΔT) based on the kinematic control error ($e_r = \omega_{des} - \omega_{actual}$). SMC is theoretically praised in academic literature for its absolute robustness against parameter variance and external disturbances—a highly desirable trait given the 400 kg to 900 kg mass fluctuation of the commercial auto-rickshaw.

However, standard first-order SMC achieves this robustness through a highly aggressive, discontinuous switching control law. This is typically modeled using a signum function:

$u(t) = -K \cdot \text{sgn}(s(t))$, where $s(t)$ represents the sliding surface defined by the error dynamics. In continuous-time mathematics, this discontinuous switching operates at an infinite frequency to keep the system states strictly bound to the sliding surface. However, when deployed in a discrete-time embedded VCU, the switching is constrained by the sampling frequency of the control loop (e.g., 100 Hz to 1 kHz). This discrete-time limitation results in a high-frequency, finite-amplitude oscillation around the sliding surface, a highly detrimental phenomenon universally known as "chattering".¹⁴

In a traditional electric vehicle architecture utilizing a central inboard motor and half-shafts, the

inherent mechanical compliance of the drivetrain—including constant velocity (CV) joints, rubber bushings, and the torsional elasticity of the axles—acts as a mechanical low-pass filter, absorbing and dampening this control chattering before it reaches the wheels. The proposed auto-rickshaw architecture, however, utilizes direct-drive independent rear hub motors. Hub motors are characterized by their rigid connection directly to the wheel hub, possessing zero mechanical compliance and exhibiting massive unsprung inertia (often adding 20 to 30 kilograms per wheel).¹⁶

When the SMC outputs a chattering, high-frequency corrective yaw moment, the local motor inverters will command violent, high-frequency electromagnetic torque ripples directly to the rigid tire contact patch. This chattering torque will induce severe longitudinal slip-stick oscillations at the tire interface, radically degrading the tire's lateral cornering stiffness and compromising the very stability the SMC is attempting to maintain.¹⁹ Furthermore, the high-frequency torque ripple is highly likely to excite the natural resonant frequency of the heavy unsprung mass (typically in the 10-15 Hz range), causing severe vertical wheel hop. This will result in catastrophic Noise, Vibration, and Harshness (NVH) propagating throughout the vehicle cabin, significantly accelerated fatigue wear on the hub motor bearings, and an unacceptable driving experience for commercial passengers.²⁰ Without the implementation of a continuous boundary layer approach or a higher-order sliding mode topology, first-order SMC is practically un-deployable on direct-drive hub motors.²³

4.2 Extended Kalman Filter Computational Burden on Automotive Microcontrollers

The Estimation module employs an Extended Kalman Filter (EKF) to clean IMU noise and fuse various sensor data streams to establish the "clean state variables" necessary for the downstream controllers. The EKF is a powerful tool, but it is notoriously computationally expensive because it must linearize non-linear system dynamics at every single timestep. This linearization is achieved via Taylor series expansion, which requires the continuous,

real-time calculation of Jacobian matrices for both the state transition model (Φ_k) and the

observation model (H_k).²⁴ Following the calculation of these Jacobians, the EKF algorithm must execute a series of dense matrix multiplications and, most critically, a highly intensive

matrix inversion to compute the Kalman Gain: $K_k = P_k H_k^T (H_k P_k H_k^T + R)^{-1}$.

Executing these matrix operations in high-precision floating-point arithmetic presents an immense computational burden for embedded systems. Commercial-grade automotive VCUs typically utilize microcontrollers such as the Infineon AURIX TC3xx family or advanced ARM Cortex-M7 series processors.²⁷ While these are highly capable, safety-rated 32-bit MCUs designed for real-time control, forcing the CPU core to execute complex matrix inversions for a multi-degree-of-freedom state vector at a 100 Hz or 200 Hz control loop frequency will monopolize the execution pipeline.²⁸

The proposed architecture explicitly specifies a strict "50ms CAN Timeout" failsafe and a "0ms bypass" for emergency torque overriding. These safety-critical functions require deterministic,

highly predictable execution timing. If the EKF matrix inversion routines cause execution pipeline stalls, cache misses, or task overruns within the Real-Time Operating System (RTOS), the primary control loop will experience severe jitter. A delayed execution loop will disrupt the transmission of the VCU watchdog heartbeat or falsely trigger the 50ms CAN timeout, leading to localized system resets or unwarranted, violent emergency vehicle shutdowns. Unless the EKF is aggressively mathematically optimized—such as by separating the covariance matrix updates into a slower background task, utilizing a sub-optimal steady-state Kalman gain, or employing U-D factorization—the standard EKF formulation is far too computationally heavy for a cost-constrained commercial VCU tasked with simultaneously managing complex torque allocation, rigorous degradation state machines, and high-speed CAN FD routing.³¹

5. Kinematic Edge Case Failures: The Front Wheel Lockup Singularity

The most critical and devastating architectural vulnerability in the proposed system resides within the Estimation Module's fundamental assumption regarding vehicle speed. The architecture explicitly states: *"Free-Rolling Front Wheel: Provides an uncorrupted, zero-slip*

reference for true longitudinal vehicle velocity (v_x)." By tethering the entire kinematic and dynamic state estimation of the vehicle to a single mechanical component, the architecture creates a catastrophic single point of failure that will inevitably trigger during the most critical driving maneuvers.

5.1 The Mechanics of Un-Driven Wheel Lockup

While the front wheel of a delta-footprint auto-rickshaw is indeed un-driven and free-rolling during steady-state cruising and acceleration, it is heavily relied upon during braking. Commercial three-wheelers typically feature hydraulic drum or basic disc brakes on the front wheel, operated via a mechanical foot pedal or handlebar lever. Because the proposed architecture explicitly assumes strict cost constraints ("commercial-grade sensors, not expensive aerospace-grade equipment"), it is highly probable that the front wheel braking system lacks an independent, high-fidelity Anti-lock Braking System (ABS), or at best, relies on a rudimentary, low-frequency hydraulic modulator.³³

In an emergency panic braking scenario—particularly on a low-friction surface such as wet tarmac, mud, or loose gravel—the driver will aggressively actuate the brake. The vehicle's mass will dynamically transfer to the front wheel (suspension dive). If the clamping force of the front brake caliper exceeds the longitudinal frictional limit of the front tire contact patch, the tire will saturate its friction circle and the front wheel will instantly lock up. At this exact millisecond, the

front wheel speed encoder will report a rotational velocity (ω_{front}) of 0 rad/s. Because the

VCU logic calculates the true longitudinal vehicle velocity as $v_x = \omega_{front} \cdot r_{tire}$, the

calculated vehicle speed v_x will instantaneously drop to 0 km/h, despite the fact that the chassis of the vehicle is still physically hurtling forward and sliding across the road surface at

high speed.³⁵

5.2 Cascading Systemic Collapse Across the VCU Logic

The instantaneous, erroneous collapse of the reference speed v_x to zero initiates a mathematically disastrous, cascading systemic failure across every interconnected control module within the VCU architecture.

1. **Gradient Handler Destruction:** The VCU calculates real-time road gradient using the equation $\theta = \arcsin((a_x - \dot{v}_x)/g)$. As the reported speed v_x drops from a cruising speed (e.g., 40 km/h) to 0 km/h within a fraction of a second, the numerical derivative $\dot{v}_x$ registers an impossibly large negative acceleration value. The mathematics of the gradient handler will interpret this massive discrepancy between the IMU acceleration and the derived encoder acceleration as the vehicle suddenly falling down a vertical cliff face. This will utterly corrupt the baseline torque requests and invalidate the Anti-Rollover control logic precisely when the vehicle requires stability.³⁶
2. **PLTR Rollover Logic Blindness:** The predictive rollover threshold equation, $PLTR \approx \frac{2 \cdot h \cdot a_y}{g \cdot B} + C_p(\dot{\delta}_f \cdot v_x)$, relies directly on v_x to calculate the dynamic penalty for rapid steering inputs (the "pothole dodge" maneuver). With the reference speed v_x reading zero, the multiplier nullifies the entire C_p term. The predictive, feed-forward capability of the rollover mitigation system is therefore entirely disabled and blinded at the exact moment the vehicle is undergoing violent, uncontrolled transient sliding dynamics.⁷
3. **Yaw Control Singularity and Spin-Out:** The most dangerous failure occurs within the Yaw Control module. The Dynamic Reference Model calculates the ideal target yaw rate using the steady-state bicycle model equation: $\omega_{des} = \frac{v_x \cdot \delta_f}{L + K_{us} \cdot v_x^2}$. If $v_x = 0$, the mathematical numerator becomes zero, and the target yaw rate (ω_{des}) immediately resolves to zero. The VCU logic concludes that the driver's intent is to travel in a perfectly straight line with zero rotation. However, the vehicle is actively sliding, and the locked front wheel provides no directional lateral control. If the vehicle begins to yaw due to uneven road friction, the IMU will register a high actual yaw rate (ω_{actual}). The Control Error is calculated as $e_r = 0 - \omega_{actual}$, resulting in a massive error magnitude. The SMC will interpret this as a catastrophic oversteer event and command a maximum corrective yaw moment (ΔT) via the independent rear hub motors to forcefully arrest the rotation. Applying massive, asymmetric torque to the rear wheels while the front wheel is locked and skidding will immediately overwhelm the remaining rear lateral grip, violently snapping the vehicle into an unrecoverable spin.⁴⁰
4. **Traction Control False Positives:** While the front wheel is locked and sliding, the

independent rear wheels may still be rotating freely. However, the VCU calculates rear slip ratio (λ) against the reference speed v_x , which is currently reading zero. The slip ratio equation will attempt to divide by zero or calculate 100% slip for both rear wheels. The active traction control will assume the rear wheels are in a state of terminal runaway slip and trigger aggressive localized torque cuts, entirely disabling the regenerative braking capability at the exact moment the driver requires maximum deceleration force.³⁴

By failing to decouple the state estimation from a single, mechanically vulnerable source, the VCU architecture guarantees a total loss of dynamic control during the most critical safety event a commercial vehicle can face: emergency braking on a compromised surface.

6. ISO 26262 Functional Safety and Degradation State Analysis

Functional safety under the ISO 26262 automotive standard mandates that electrical and electronic (E/E) systems must strictly mitigate unreasonable risk due to hazards caused by malfunctioning behavior. The standard requires continuous fault detection, the immediate transition to a highly defined safe state, and the implementation of robust fault tolerance mechanisms.⁴⁵ The VCU's proposed degradation state machines and hardware failsafe mechanisms present severe, legally actionable safety violations when applied to a delta-footprint vehicle.

6.1 Evaluation of Proposed Software Degradation States

The Plausibility & Degradation Manager utilizes analytical redundancy—comparing virtual yaw derived from the rear wheel speed delta against the actual IMU yaw rate—to detect sensor drift. While this aligns with standard ASIL decomposition and signal vetting practices, the subsequent fallback states defined by the architecture are highly problematic.

Fallback State	Proposed Action	ISO 26262 Vulnerability Analysis
State 1: IMU Loss	Open-loop vectoring, speed capped at 25 km/h.	Open-loop vectoring relies purely on steering angle without feedback. On μ -split surfaces, open-loop torque will induce uncontrollable yaw. Capping speed at 25 km/h does not negate torque steer physics; the vehicle will spin out.
State 2: Encoder Loss	Disable e-ABS, limit torque to 50%.	Disabling slip control while allowing 50% torque output

		does not constitute a "safe state." Furthermore, losing encoders destroys the v_x reference, leading to the cascading kinematic collapses detailed in Section 5.
State 3: Steering Loss	Enforce Virtual Locked Axle ($T_L =$).	A delta trike with a locked rear axle possesses massive, dominating understeer. If the steering fails while mechanically deflected in a turn, locking the rear axle will cause the front wheel to scrub and plow straight ahead, forcing the vehicle out of its lane and into oncoming traffic. ⁴⁵
State 4: CAN Timeout	Execute 0.2g stop, trigger EPB.	Engaging an Electronic Parking Brake (EPB) dynamically on a three-wheeler risks locking the rear wheels, leading to immediate lateral instability and unrecoverable oversteer.
State 5: Hydraulic Leak	Maximize regenerative braking for "Virtual ABS."	Regenerative braking is strictly constrained by electrochemical battery limits. If the battery State of Charge (SOC) is 100%, or the ambient temperature is near freezing, the Battery Management System (BMS) will strictly refuse charge current to prevent lithium plating. The VCU will command regen, the BMS will block it, and the vehicle will possess absolutely zero stopping power. ⁴⁸

6.2 The ASIL D Hardware Failsafe Violation: Contactors and Resistor Banks

The most egregious functional safety violation resides in the architecture's "Total System Collapse" fallback (State 7). The proposal dictates an ASIL D hardware bypass wherein an

E-Stop activation or a lost VCU watchdog heartbeat directly fires a mechanical High-Voltage Contactor to disconnect the battery and drop the spinning motor phases across a passive Dynamic Braking Resistor (DBR) Bank.⁵⁰

While dynamic braking resistors are common in industrial conveyor or elevator drives, deploying them via high-voltage contactors as a primary ASIL D automotive failsafe introduces multiple catastrophic points of failure:

1. **Electromechanical Latency:** High-voltage contactors are inherently mechanical devices, relying on physical return springs and electromagnetic coils to physically separate massive copper busbars. The typical make/break time for an automotive high-voltage contactor is between 20 milliseconds and 50 milliseconds. In a total loss-of-control emergency scenario, an uncontrolled 50ms delay before deceleration torque is applied is an unacceptable hazard.
2. **Contact Welding:** In the event of a high-speed VCU failure, the dual rear hub motors will be spinning at maximum RPM, acting as powerful generators producing massive back-electromotive force (BEMF). Dropping this unconstrained BEMF across a low-impedance passive resistor bank will induce an immediate, massive transient current spike. If this transient current exceeds the maximum breaking capacity of the contactor, the resulting electrical arc will physically melt and weld the contactor pads shut. Once welded, the system cannot be disconnected, the fault cannot be cleared, and a permanent thermal runaway risk is established.⁵²
3. **Resistor Thermal Overload and Fire Hazard:** If the fault occurs at the vehicle's top speed while descending a steep, prolonged gradient, the kinetic energy of the 900 kg vehicle must be entirely converted into thermal energy by the DBR. The energy dumped into the resistors will rapidly exceed their thermal mass design limits. Without active, high-volume liquid cooling, the resistor bank will rapidly overheat, potentially melting surrounding wiring harnesses or initiating a catastrophic vehicle fire, directly violating ISO 26262 directives concerning the prevention of thermal events.⁵³

For high-voltage Permanent Magnet Synchronous Motors (PMSMs) utilized in automotive EV traction systems, the globally recognized industry-standard ASIL D safe state is not a mechanical resistor bank, but an electronic **Active Short Circuit (ASC)**.⁵⁶

7. Concrete Engineering Solutions and Architectural Remediation

To elevate the proposed VCU architecture from a theoretically flawed concept to a state of commercial readiness and strict ISO 26262 compliance, the logic and hardware topography require extensive, foundational remediation. The following patches apply industry-standard automotive control and functional safety engineering practices to systematically seal the identified vulnerabilities.

7.1 Geometric Correction of the Predictive Lateral Load Transfer Ratio

The PLTR equation must be entirely refactored to respect the unique asymmetric geometry of

the delta footprint. The control logic must establish the true tipping axis, which is defined by the angle α relative to the longitudinal centerline. This angle is calculated geometrically as $\alpha = \arctan\left(\frac{B/2}{L}\right)$.

The static half-track width ($B/2$) must be completely replaced by the orthogonal distance from the center of gravity to this new tipping axis (d_{tip}). Crucially, the dynamic PLTR equation must vectorially project both the lateral acceleration (a_y) and the longitudinal acceleration (a_x) onto this diagonal axis to capture combined cornering and braking forces accurately. Furthermore, to account for the massive 400 kg to 900 kg payload variance, the VCU must integrate a dynamic payload estimator. This can be achieved using low-cost suspension deflection sensors (potentiometers) or by implementing an extended mass-estimation observer within the EKF to continuously update the estimated CG height (h) and longitudinal position (l_f) in real-time.

The corrected, physics-based overturning moment constraint for the delta trike becomes:

$$PLTR_{delta} = \frac{h}{g} \frac{a_y \cos \alpha + a_x \sin \alpha}{d_{tip}} + C_p \delta_f v_x$$

This fundamental geometric correction ensures that combined cornering and braking forces map accurately to the triangular footprint, eliminating false-positive interventions when unloaded and preventing catastrophic missed rollover predictions when the auto-rickshaw is carrying maximum passenger capacity.

7.2 Higher-Order Sliding Mode Control and Boundary Layer Saturation

To eliminate the destructive mechanical chattering inherent in the proposed first-order Sliding Mode Controller while preserving its highly desirable robustness to mass variance, the Yaw Control module must transition to a more sophisticated switching topology. The architecture must implement either a **Boundary Layer Saturation** function or transition entirely to a **Higher-Order Sliding Mode (HOSM)** controller, specifically employing the **Super-Twisting Algorithm (STA)**.²³

If retaining first-order SMC, the discontinuous signum function $u(t) = -K \cdot \text{sgn}(s)$ must be replaced with a continuous boundary layer approach:

$$u(t) = -K \cdot \text{sat}\left(\frac{s}{\Phi}\right)$$

where the parameter Φ defines the calibrated thickness of the boundary layer around the sliding surface. Within this specific boundary layer, the control response transitions from

discrete switching to a linear, continuous mathematical response, effectively acting as a high-gain, low-pass filter on the corrective yaw torque demand.⁶¹ This critical modification eliminates the high-frequency limit cycle switching before it reaches the motor inverters, smoothly blending the torque request and completely preventing the excitation of the hub motor's heavy unsprung mass natural frequencies. This preserves passenger ride comfort, prevents premature spline and bearing fatigue, and maximizes the tire's longitudinal traction limit.

7.3 Kinematic-Dynamic Hybrid Observers for Resilient State Estimation

To resolve the catastrophic edge case of front-wheel lockup resulting in total systemic collapse, the Estimation Module must fundamentally decouple the vehicle speed reference (v_x) from pure, blind reliance on the un-driven front wheel speed encoder. The Extended Kalman Filter must be deeply reconfigured to implement a **Kinematic-Dynamic Hybrid Observer**.³⁴ The estimation system must continuously cross-reference three distinct data streams: the front wheel speed (v_{front}), the integrated longitudinal acceleration from the IMU ($a_x dt$), and the aggregated average speed of the driven rear wheels ($\frac{v_{RL} + v_{RR}}{2}$). A dedicated slip-detection logic layer must be introduced into the EKF pre-processing phase. If the calculated deceleration rate of the front wheel ($\dot{v}_{front}$) suddenly exceeds a physically plausible threshold (e.g., decelerating faster than $-1.2g$), the VCU logic must immediately deduce that the front wheel has locked up and is sliding.³⁸ Upon this trigger condition, the EKF must dynamically drop the observation weighting matrix associated with the front wheel to zero, actively ignoring the corrupted 0 km/h data. The EKF then seamlessly transitions to dead-reckoning state estimation via IMU acceleration integration, continuously blended and corrected by the rotating rear wheel speeds. This hybrid observation strategy guarantees that the critical state variable v_x remains a highly accurate, resilient reflection of true chassis speed during violent emergency braking, maintaining the functional integrity of the Yaw Control, Traction Control, and PLTR modules when they are needed most.

7.4 Active Short Circuit Implementation for ASIL D Compliance

The utilization of a mechanical High-Voltage Contactor and a passive Dynamic Braking Resistor Bank must be entirely expunged from the safety architecture. To achieve true ISO 26262 ASIL D compliance, the system must employ an isolated **Active Short Circuit (ASC)** implementation. This safety response must be managed not by the primary VCU processing core, but by a dedicated, ASIL D-rated safety companion microcontroller—such as an independent lockstep core or a heavily isolated external hardware watchdog supervisor.⁶³

Feature	Dynamic Braking Resistor (Proposed)	Active Short Circuit (Recommended Solution)

Actuation Mechanism	Mechanical High-Voltage Contactor	Direct electronic trigger to Inverter Gate Drivers
System Latency	20 ms – 50 ms (Mechanical lag)	< 1 ms (Solid-state electronic switching)
Primary Failure Modes	Contactor arc welding, Resistor fire	Minimal (requires only intact 12V gate drive power)
Regulatory Alignment	Legacy industrial/elevator design	Automotive ASIL D standard for PMSM EV traction

Upon the detection of a lost heartbeat, severe CAN timeout, or manual E-Stop activation, the safety watchdog hardware directly overrides the software stack and forces the gate drivers of the local motor inverters to turn on all three high-side IGBTs (or, alternatively, all three low-side IGBTs) simultaneously.⁵⁷ This action creates an immediate, symmetrical electrical short circuit within the motor's stator windings.

As the vehicle continues to move, the spinning permanent magnets in the hub motors generate a massive back-EMF that drives a high transient current through this internal short circuit. Crucially, this current is strictly and safely limited by the stator's own internal inductance and resistance, generating massive electromagnetic drag.⁵⁶ The ASC topography provides incredibly rapid, highly reliable, purely physics-based deceleration torque that is entirely decoupled from the VCU software stack. It requires no mechanical relays or switching devices, handles 100% battery SOC scenarios flawlessly because the energy is dissipated internally as heat within the massive thermal sink of the motor stators, and completely eliminates the profound fire hazards associated with external resistor banks.

8. Conclusion

The proposed software-defined VCU architecture demonstrates a progressive, forward-thinking approach to managing the exceedingly complex dynamics of independent rear hub motors on a commercial three-wheeled platform. The ambition to replace heavy mechanical differentials with intelligent torque vectoring is sound, and the integration of predictive rollover models showcases a mature understanding of active safety requirements. However, the current iteration of the architecture is severely hindered by the direct, unadjusted porting of four-wheeled mathematical models onto a delta footprint, a naive over-reliance on the idealized performance of commercial-grade sensors, and the deployment of continuous-time control theories that translate poorly to the discrete, high-unsprung-mass realities of hub motor hardware. The most critical vulnerabilities identified in this review—specifically the total collapse of the kinematic state estimation matrix upon

front-wheel lockup, the unconstrained yaw generation during μ -split traction control, and the catastrophic fire and welding hazards of the mechanical DBR failsafe—pose severe, unacceptable risks to vehicle occupants and surrounding traffic.

By systematically refactoring the PLTR equation to map the true diagonal tipping axis, implementing continuous boundary layers in the Sliding Mode Controller to suppress

destructive NVH chattering, moving to a Kinematic-Dynamic Hybrid Observer for resilient vehicle speed estimation, and replacing the mechanical hardware bypass with an ASIL D-compliant Active Short Circuit topography, the architecture can achieve both unparalleled commercial performance and strict regulatory compliance. Implementing these industry-standard solutions will yield a Vehicle Control Unit capable of delivering a safe, stable, and highly predictable driving experience, successfully taming the complex dynamics of the delta-footprint auto-rickshaw even under the harshest commercial operating constraints.

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